

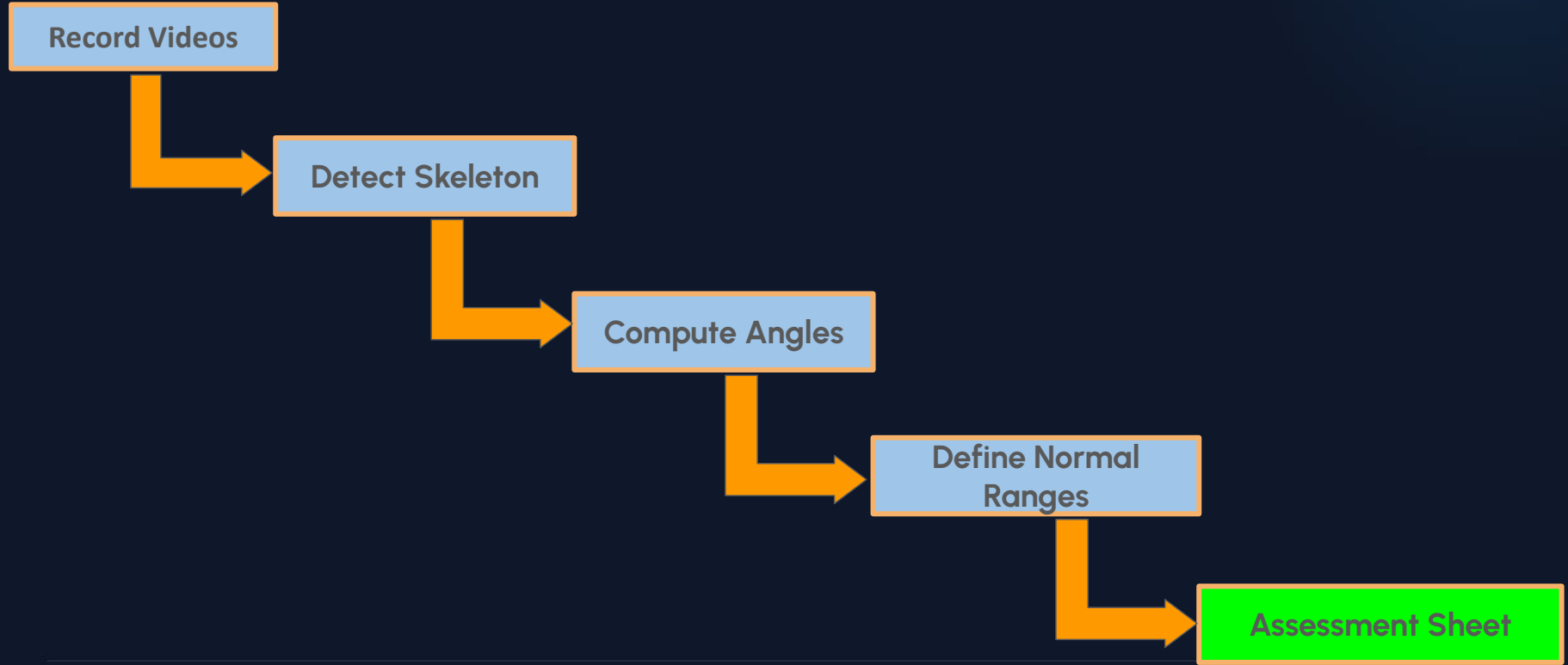
Automated Orthopedic Functional Assessment

AI-Based 3D Human Pose Estimation for Automated Clinical ROM Analysis

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Current Workflow



1. Record Assessment Videos

Recorded Movements

Data Collection Setup

- Smartphone camera
- Markerless recording
- Indoor environment
- Single subject
- Each Movement recorded twice

Recorded Movement Categories

Spinal Kinematics

- Lateral Bending (Left/Right)
- Axial Rotation (Left/Right)

Lower Extremity

- Knee Flexion and Extension

Distal Joints

- Ankle Dorsiflexion
- Ankle Plantarflexion
- Ankle Rotation

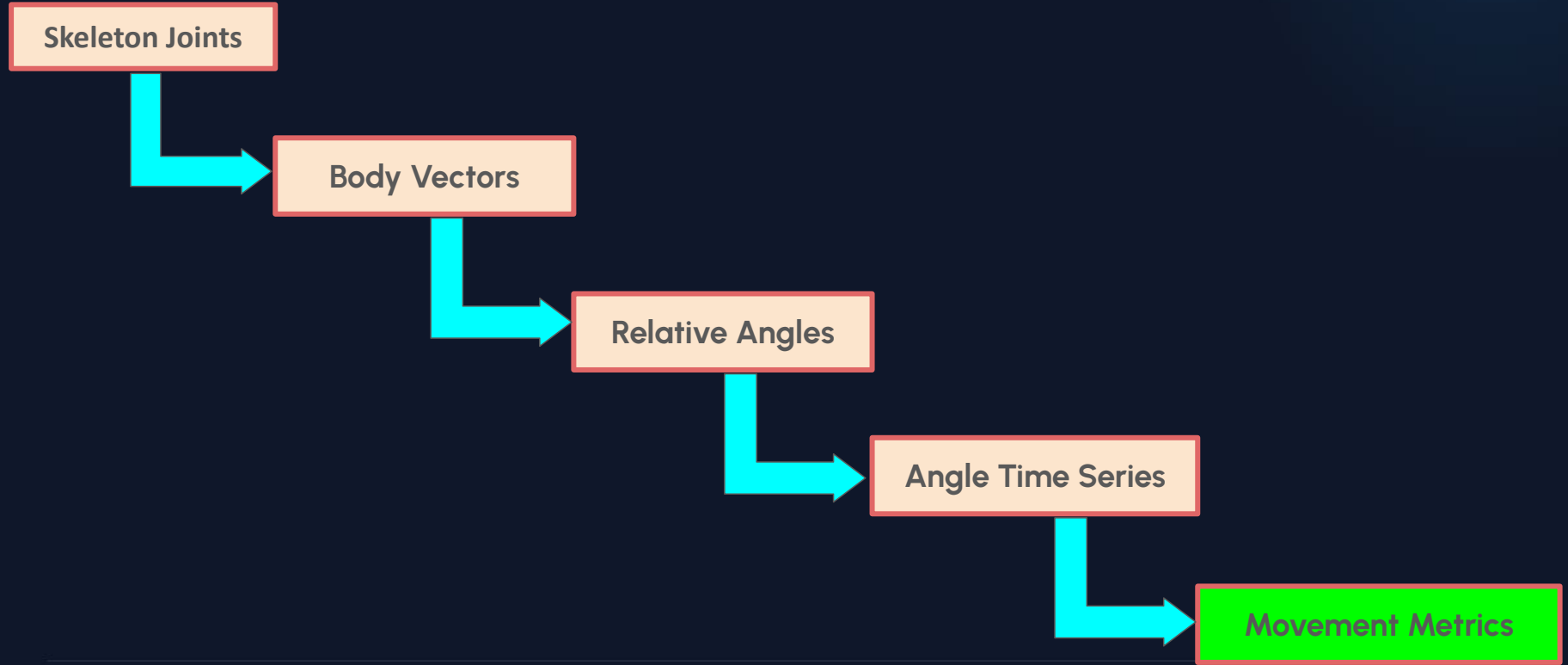
2. Skeleton Detection

Detecting Skeleton

Model	Output	Notes
MediaPipe	33 landmarks	Real-time 2.5D baseline, rapid temporal initialization
MeTRAbs	3D skeleton	3D motion and depth analysis
RTMPose	2D skeleton	High-precision 2D baseline
MoveNet	17 landmarks (2D)	Fast, excellent temporal tracking consistency

3. Computing Angles

From Skeleton to Orthopedic Measurements



From Skeleton to Orthopedic Measurements

Stage	Input	Output
Vector Construction	Joint Coordinates	Body Vectors
Relative Angle Computation	Body Vectors	Biomechanical Angles
Time Analysis	Angle Sequence	ROM, Symmetry, Peak Angle

From Skeleton to Orthopedic Measurements

Movement	Joints	Measurement
Back Rotation	Shoulders + Hips	Torso twist angle
Back Bending	Shoulder center + Hip center	Flexion angle
Side Bending	Spine landmarks	Lateral bending angle
Knee Movement	Hip-Knee-Ankle	Knee flexion angle
Ankle Movement	Knee-Ankle-Toe	Ankle angle

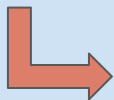
Example: Back Rotation

Joints:

- Left Shoulder - Right Shoulder
- Left Hip - Right Hip

Vectors:

- $V_{\text{shoulder}} = \text{Right_Shoulder} - \text{Left_Shoulder}$
- $V_{\text{hip}} = \text{Right_Hip} - \text{Left_Hip}$



Angle:

$$\theta_{\text{rotation}} = \cos^{-1} \left(\frac{v_{\text{shoulder}} \cdot v_{\text{hip}}}{||v_{\text{shoulder}}|| \times ||v_{\text{hip}}||} \right)$$

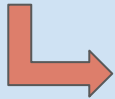
Example: Back Bending

Joints:

- Left Shoulder - Right Shoulder
- Left Hip - Right Hip

Vector:

- $V_{spine} = \text{Shoulder_Center} - \text{Hip_Center}$



Angle:

$$\theta_{bend} = \cos^{-1} \left(\frac{v_{spine} \cdot v_{vertical}}{\|v_{spine}\| \|v_{vertical}\|} \right)$$

Time Analysis

Angle Time Series:

- Frame 1 → 5°
- Frame 2 → 12°
- Frame 3 → 20°
- Frame 4 → 35°
- Frame 5 → 42°

Extract:

- Maximum angle
- Range of Motion (ROM)
- Left/right symmetry
- Movement smoothness

Movement Metrics

Metric	Value
Forward Flexion	42°
Left Rotation	28°
Right Rotation	31°
Side Bend Left	18°
Side Bend Right	21°

4. Define Normal Movements

Defining Normal vs Abnormal Movement

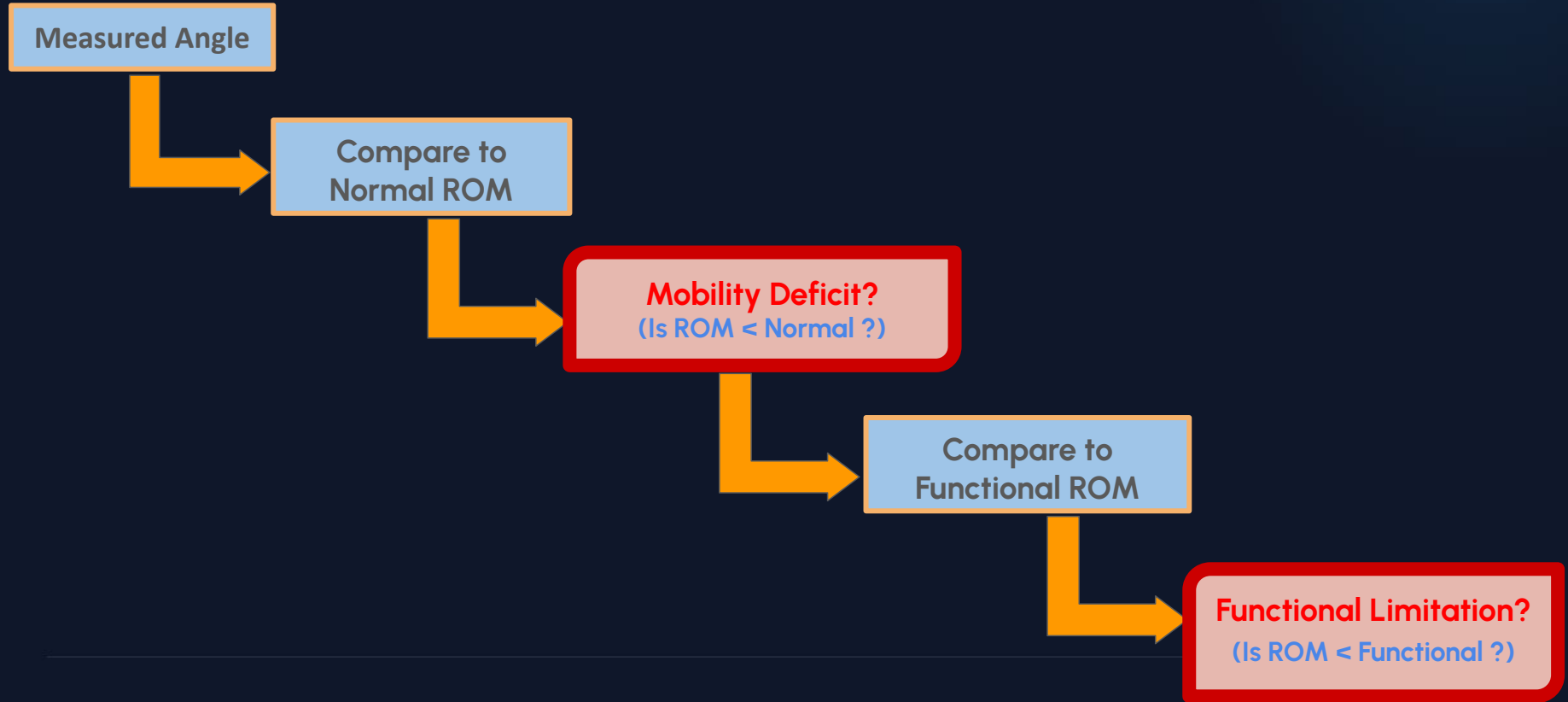
Layer 1: Normal ROM

- The maximum joint movement
- Reference: AAOS Joint Motion (American Academy of Orthopaedic Surgeons guidelines)
- Purpose: Detect mobility limitations
- Example:
 - Measured Flexion = 40°
 - Reference Flexion = 60°

Layer 2: Functional ROM

- The minimum joint movement required to perform everyday activities.
- Reference: standard Activities of Daily Living (ADL)
- Purpose: Estimate functional impact
- Example:
 - Can the measured ROM support walking, bending, reaching, etc.?

Defining Normal vs Abnormal Movement



5. Assessment Sheet

Assessment Variables

- Core Metrics:

Variable	Purpose
Flexion Angle	Quantify forward bending
Rotation Angle	Quantify trunk rotation
Side-Bending Angle	Quantify lateral bending
ROM Achievement (%)	Compare to expected ROM
ROM Deficit (°)	Mobility loss
ROM Symmetry	Left–right comparison

Assessment Variables

- Advanced Metrics:

Variable	Purpose
Pelvic Stability	Detect compensation by pelvis
Planar Deviation	Detect unintended movement planes
Kinematic Jitter	Assess movement smoothness

Possible Findings

Mobility

- Normal ROM
- Reduced ROM
- Severely Reduced ROM

Symmetry

- Symmetric Movement
- Mild Asymmetry
- Significant Asymmetry

Functional Status

- Functional ROM Preserved
- Functional Limitation Detected

Movement Quality

- Stable Movement
- Pelvic Compensation Detected
- Excessive Planar Deviation
- High Kinematic Jitter